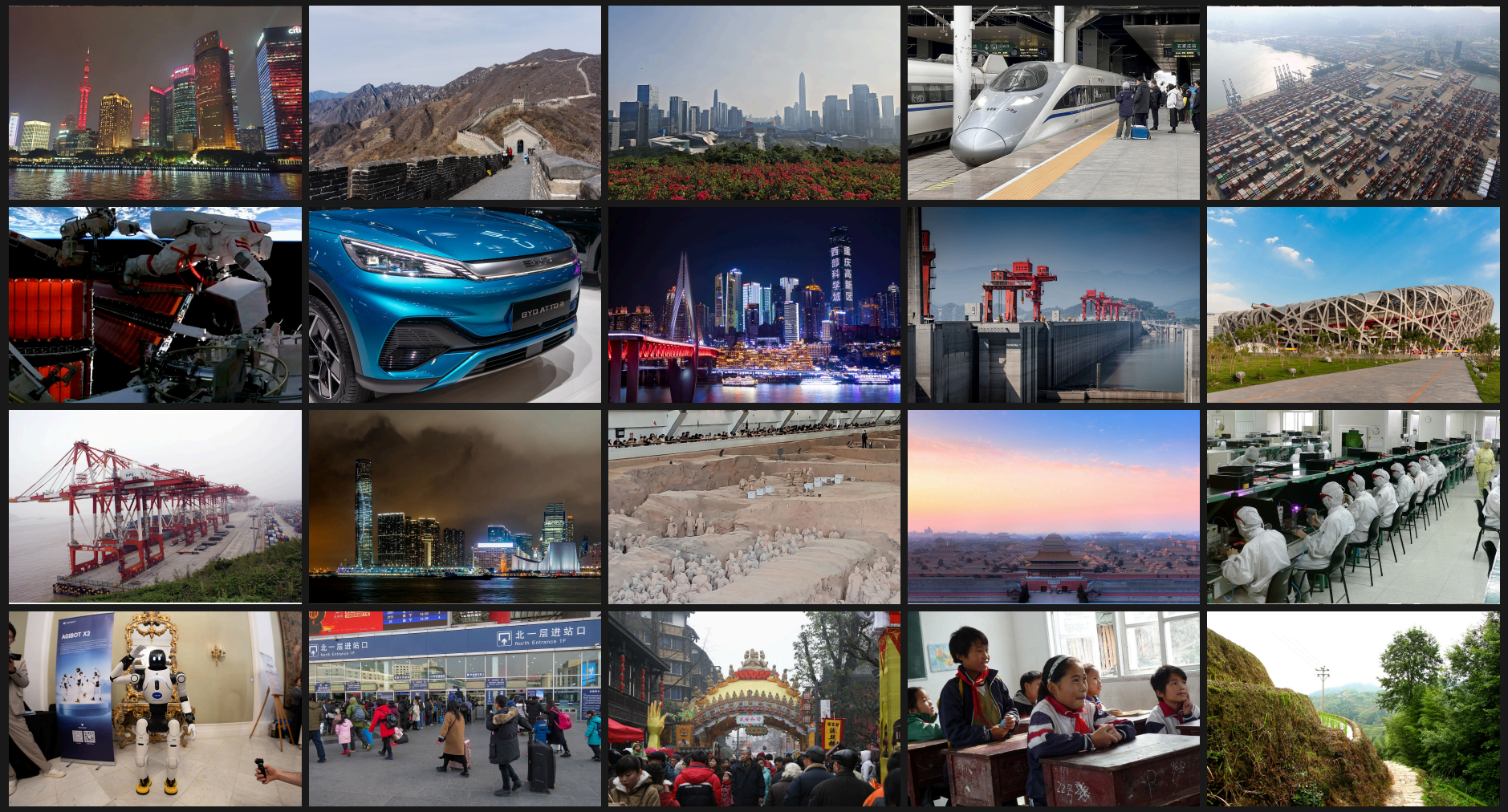
THE CHINESE ECONOMY

A JOURNEY INTO HISTORY AND ECONOMICS

Lecture 1: Foundations


Political
Physical



20 images. Credits and licences: course website, Credits page

History

"…the science of men in time."

Marc Bloch

we will use it

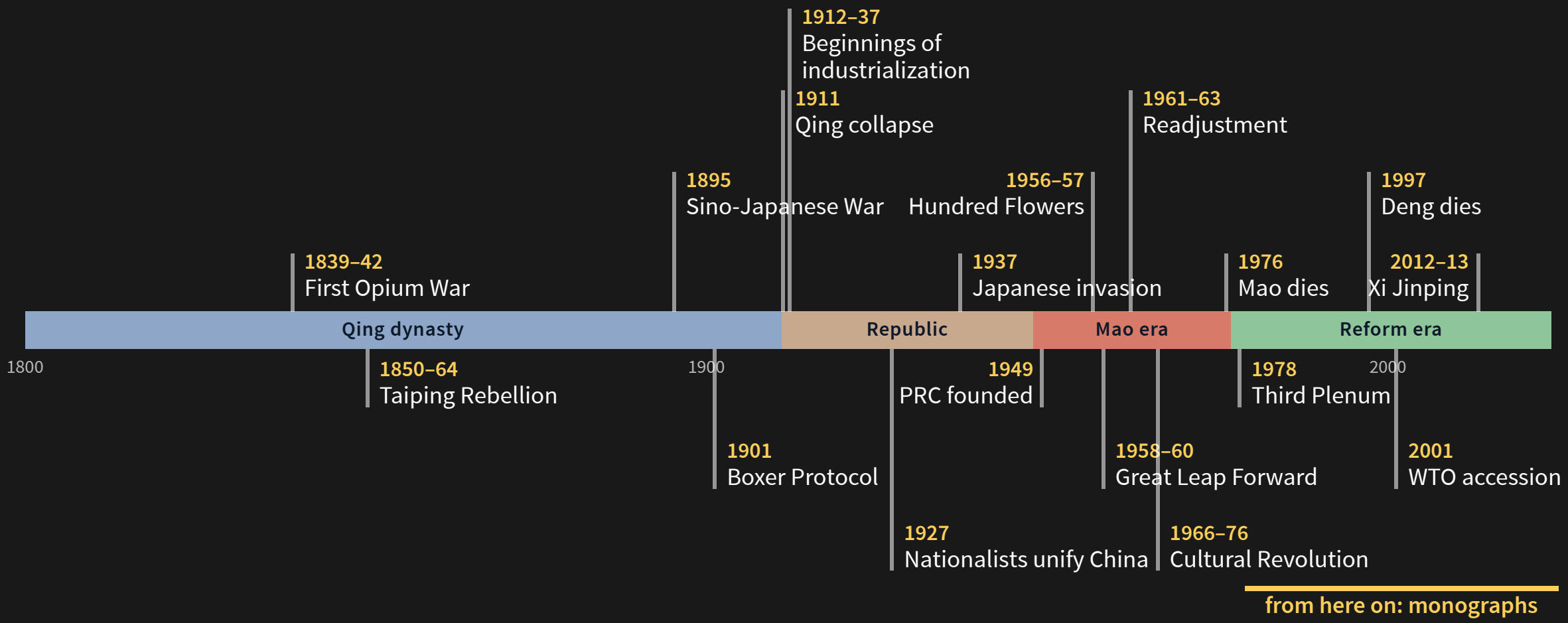
Economics

"…the science which studies human behaviour as a relationship between ends and scarce means which have alternative uses."

Lionel Robbins, 1932

we will do it

150 YEARS IN ONE LINE



THE COURSE IN SEVEN BLOCKS

- **Foundations** — lecture 1 · history and institutions, from 1800 to 1978 *(today)*
- **Causality** — lectures 2-3 · the method: what a cause is, and how economists find one
- **Reallocation across sectors** — lectures 4-5 · from the farm to the factory: GDP, hukou, migration
- **Reallocation within sectors** — lecture 6 · firms, state-owned enterprises, misallocation
- **Trade** — lectures 7-8 · growth, the WTO, tariffs and trade wars
- **Innovation** — lectures 9-11 · China at the Frontier
- **Conclusions** — lecture 12

RULES OF THE JOURNEY

- **Website** — syllabus and slides: **framana111.github.io/the-chinese-economy**
- **Exercises** — from time to time, a short problem to take home: tools to learn the logic of economics, analysis and reflection; **not graded**
- **Assessment** — final written exam

A QUICK POLL

Yardsticks

Average annual **real GDP growth, 2005-2024**

- **Italy** 0.3% a year
- **European Union** 1.3% a year
- **United Kingdom** 1.4% a year
- **United States** 2.1% a year
- **Japan** 0.6% a year

CHINA, 2005-2024

7.9% a year — real GDP, 20 years

- The economy became **4.6 times** larger (+356%)
- Italy +5% · European Union +30% · United Kingdom +30% · United States +51% · Japan +12%

POLL 2

What drove this growth? Rank all six, from 1 (most important) to 6 (least)

- movement of workers from the countryside to the cities
- reform of state enterprises and better allocation across firms
- foreign trade and WTO entry
- innovation and technology catch-up
- high investment and saving
- industrial policy and state direction



scan with your phone, or open:
docs.google.com/forms/d/e/1FAIpQLSdHRvs0LmjfFJTO84sf_KCfSaEThs8GHtHyVPeH-j882IEStg/viewform


Aihui
West: 6% of the population
Tengchong
East: 94%

CHINA, AROUND 1800

- About 381 million people in 1820 — around **36% of the world**
- About a third of world GDP (32%, Maddison); living standards close to the world average
- Over 90% rural: very high yields per hectare, low output per worker
- A dense, commercial economy: markets, money, contracts, competition — small and household-based ("bottom-heavy")

…ALREADY UNDER STRAIN

- **Land**: nearly all cultivable land in use by 1800 — a "high-level equilibrium trap" (Elvin 1973)
- **State**: a thin layer — revenue 2-3% of national income; one official per 32,000 people (Europe: one per 600-800)
- **Inequality**: the gentry (under 1% of people) took about 20% of national income
- "The Chinese peasant is like a man standing on tiptoe up to his nose in water" (Tawney)



Johanna Helena Herolt, *A Poppy in Three Stages of Flowering, with a Caterpillar, Pupa and Butterfly*, late 17th-early 18th century.
Metropolitan Museum of Art (CC0)

TRADE, SILVER AND OPIUM



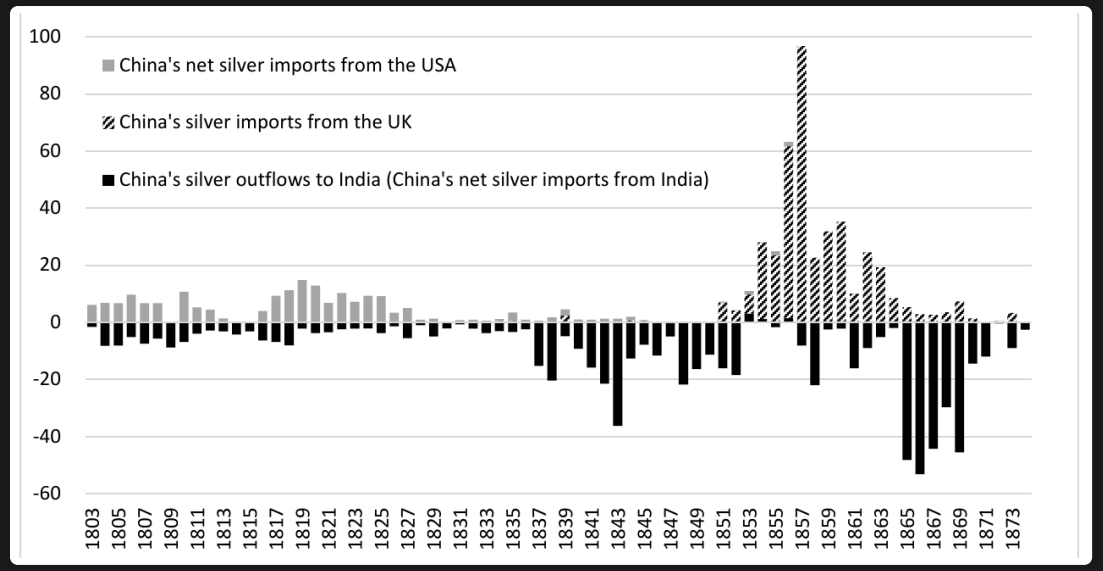


The opium fleet on the Ganges, c. 1850. Lithograph after W. S. Sherwill (Wellcome, CC BY 4.0)

- For centuries China sold tea, silk and porcelain, and received **silver**
- Britain needed something China would buy: **opium**, grown in India and sold at Canton by private British traders. Its proceeds paid for the tea
- By the 1830s China imported more than it exported

SILVER AND OPIUM: WHAT THE SERIES SHOW

- Silver left China for India in the late 1830s and the 1840s (black bars below zero)
- Then the 1850s: a very large inflow from Britain. And in the 1860s, a new outflow
- Opium shipped from India roughly doubled in five years: **21,885 chests** (1834/35) to **40,200** (1838/39)
- Did opium push the silver out — or did the scarcity of silver push opium in? Historians disagree: Irigoin and co-authors find only a weak association; Lin (2006) told a supply-side story
- Two lines that move together can mean nothing, and can run either way: spurious correlations



Silver trade of China with India, Britain and the US, 1803-1874 (million rupees). Irigoin et al. (2023), Fig. 2

WHAT THE OPIUM SURGE DID



Two wealthy Chinese opium smokers. Gouache on rice paper, 19th century (Wellcome Collection, CC BY 4.0)

- Imports of Indian opium doubled in five years: 21,885 chests (1834/35) to 40,200 (1838/39) — *Greenberg, via Farooqui (2016)*
- More users, but no reliable count: estimates run from a few million to over ten million — *MIT Visualizing Cultures*
- It reached officials, soldiers and the court; local officials took bribes to ignore the ban — *MIT Visualizing Cultures*
- The court split between prohibition and legalisation — *Polachek (1992)*
- Silver outflow and currency stress: usually blamed on opium, a link recent work questions — *Lin (2006); Irigoin et al.*

THE OPIUM WARS

- **1839-42** — China tries to stop the opium inflow; Britain attacks. *Treaty of Nanjing* (1842): Hong Kong ceded, five treaty ports opened, indemnity of 21 million silver dollars, end of the Cohong monopoly
- **1856-60** — second war; treaties of Tianjin (1858) and Convention of Peking (1860): more ports, envoys in Beijing, missionaries, opium legalised
- Through 1895, **three more** wars, all lost: reparations, more treaty ports (over 80 at the peak)



The *Nemesis* destroying Chinese war junks, Anson's Bay, 7 January 1841. Print by Edward Duncan, 1843 (public domain)

FROM BRITAIN TO JAPAN

- **1895**: Sino-Japanese War → Taiwan to Japan; the Treaty of Shimonoseki lets foreigners invest — modern industry begins
- **1901**: after the Boxer Rebellion, eight powers occupy Beijing; Boxer Protocol: indemnity, tariff revenue under foreign control
- **1911**: the Qing dynasty collapses
- Share of world GDP: **32% (1820) → 9% (1913)**



China, the cake of kings and... emperors. *Le Petit Journal*, January 1898 (public domain)



The Battle of the Yalu River, 17 September 1894. Kobayashi Kiyochika (British Library, public domain)

1912-1937: FIRST INDUSTRIALIZATION

- After 1911: warlords; **1927** the Nationalists unify the country; the "Nanjing decade"
- Modern factories grow **8-9% a year** — from 2% of GDP and 0.4% of the labour force
- Two patterns: **Treaty-Port** industry (light, Chinese-owned by the 1930s) and **Manchuria** (Japanese, heavy, few spillovers)

1937-1949: WAR AND CIVIL WAR

- **1931** Manchukuo; **1937** Japan invades China proper — a decade of war
- Manchuria's industry grows 14% a year (1936-42); Shanghai never regains its 1936 output before 1949
- The Nationalist state runs the war economy; **hyperinflation**: prices in 1948 = 660 million (1936 Shanghai = 100)
- 1949: the Communist victory

WHERE THIS LECTURE GOES

- **1800 → 1949** — how a great economy arrived at the revolution
- **1949 → 1978** — a planned economy: how it worked, what it did, and the Great Leap Forward
- **1978** — the turn, and a question we cannot yet answer

THE ECONOMIC PROBLEM

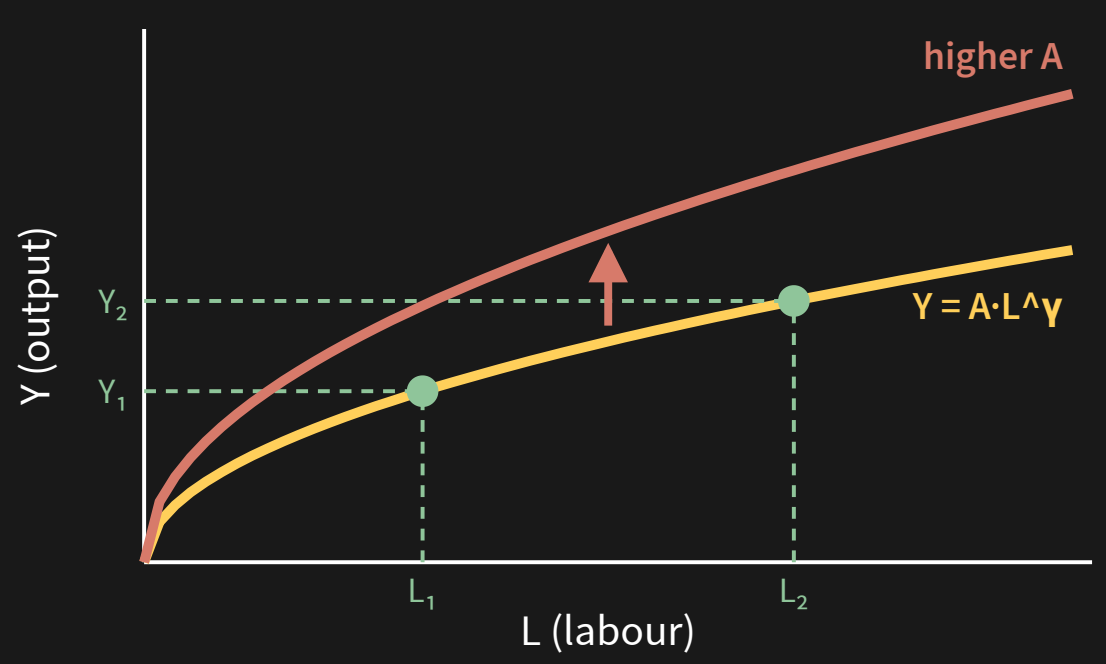
Who gives what to whom — and who decides?

- Millions of different actors: households, farms, factories
- Each must decide how much to take and how much to give to the others
- Two ways to coordinate: through **prices**, or through **quantities** (a plan)

WHAT DOES A PRICE DO?

- It says how **scarce** something is, relative to everything else
- It carries information nobody holds in full — *Hayek (1945)*: knowledge is dispersed; prices coordinate separate actions
- It rewards whoever economises or finds a substitute
- *Other traditions: prices also reflect technology and the distribution of income (Ricardo, Marx, Sraffa)*

HOW PRODUCTION WORKS



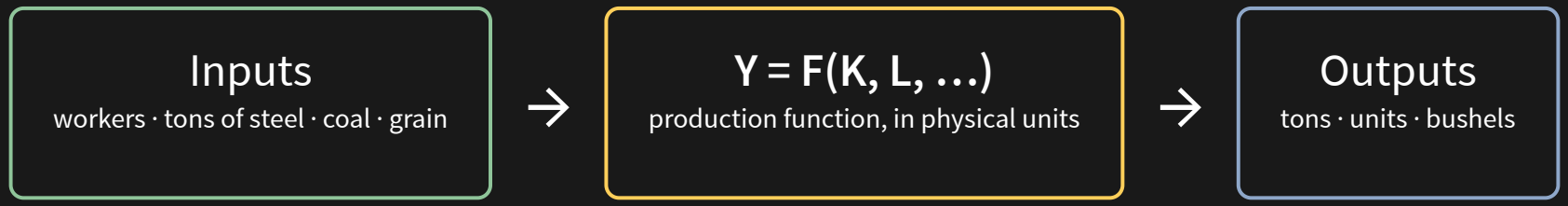
$Y = A \cdot L^{\gamma}$, with $0 < \gamma < 1$

Y output · **L** labour, the only input for now · **A** productivity

More L: we move **along** the curve, to a higher Y. Each extra worker adds less: diminishing returns

Higher A: the **whole curve** shifts up. Same labour, more output

A PLAN WORKS ON QUANTITIES



- The planner sets **targets** and allocates **quantities** to each producer
- Prices are still there — but set by the state, not by the market

CHINA'S COMMAND ECONOMY

- Initially inspired by the **USSR model**, but with differences
- The state owns the big factories; collectives own the land
- **Commands**: production targets; resources allocated directly — prices lose their role as the main signal
- The state still *sets* prices — industry dear, agriculture cheap — to channel resources to itself; compulsory grain procurement from 1953
- The Party controls managers' careers
- Labour fixed in place: tight limits on movement, jobs for life
- **Main limitations**
    - **Weak control**: weaker than in the USSR (Naughton). Symbol: Beijing could allocate at most 600 product varieties; the USSR, 60,000 by the 1970s

MEASURING WITHOUT MARKET PRICES

- GDP adds up value at market prices — what is "value added" without them?
- China used the **Material Product System (MPS)** until 1992: it counts material production (agriculture, industry, construction, freight transport, distribution) but not *non-material* services (education, health, finance, housing, administration)
- MPS and GDP coexisted in 1985-92; GDP only from 1993
- So we look at a different indicator, the planners' main lever: **investment**

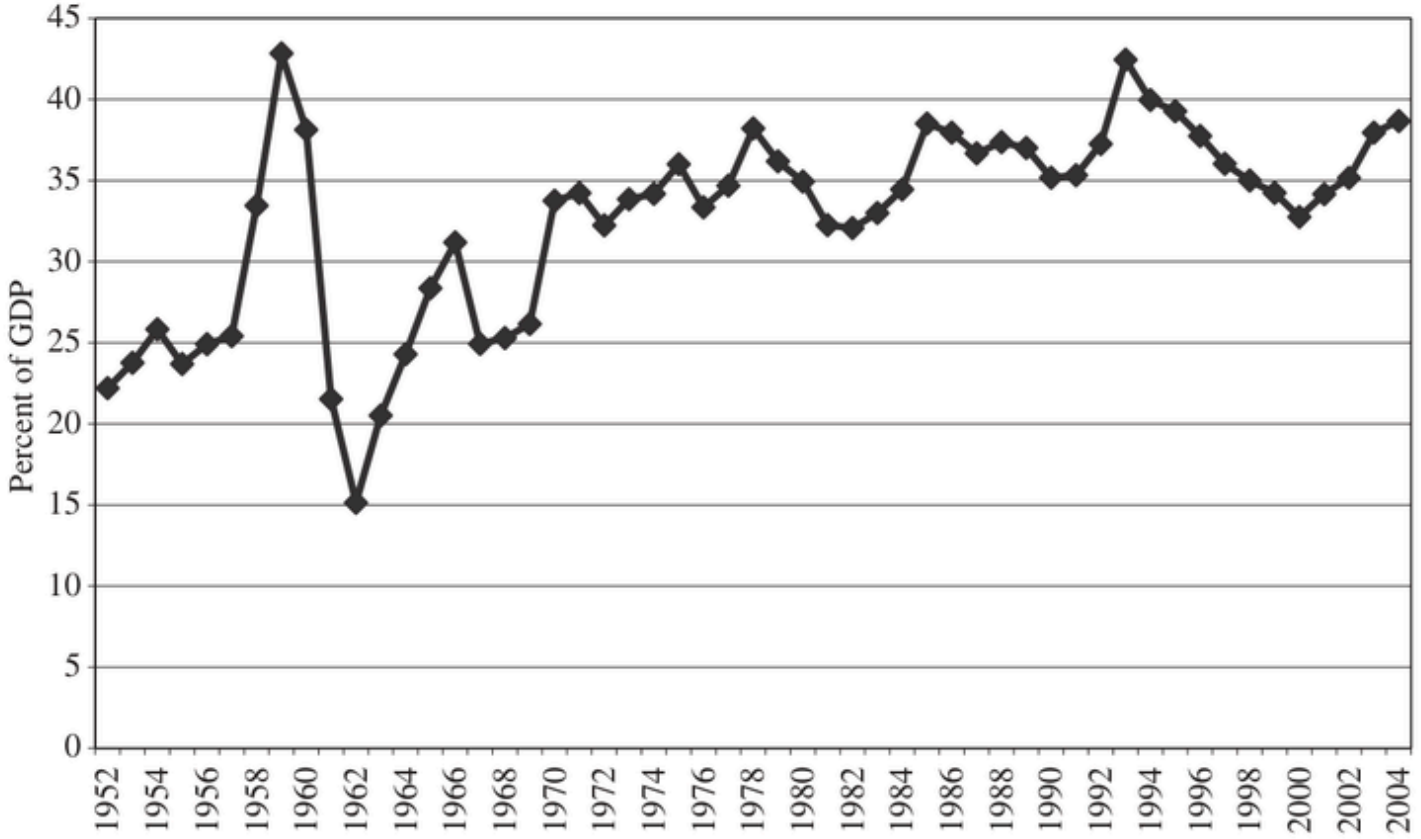


Figure 3.1
Investment as a share of GDP

Naughton (2007), *The Chinese Economy*, Fig. 3.1

LEAPS AND RETRENCHMENTS

- A **leap**: investment grows more than 20% in a year — rapid growth *and* political mobilisation; radical, "leftist" phases
- Then **retrenchment**: slower investment, consolidation, often a more moderate climate (Hundred Flowers 1956-57; readjustment 1961-63; drift 1972-76)
- "Policy swings like a pendulum from left to right and back again" — and drives, and is driven by, economic developments

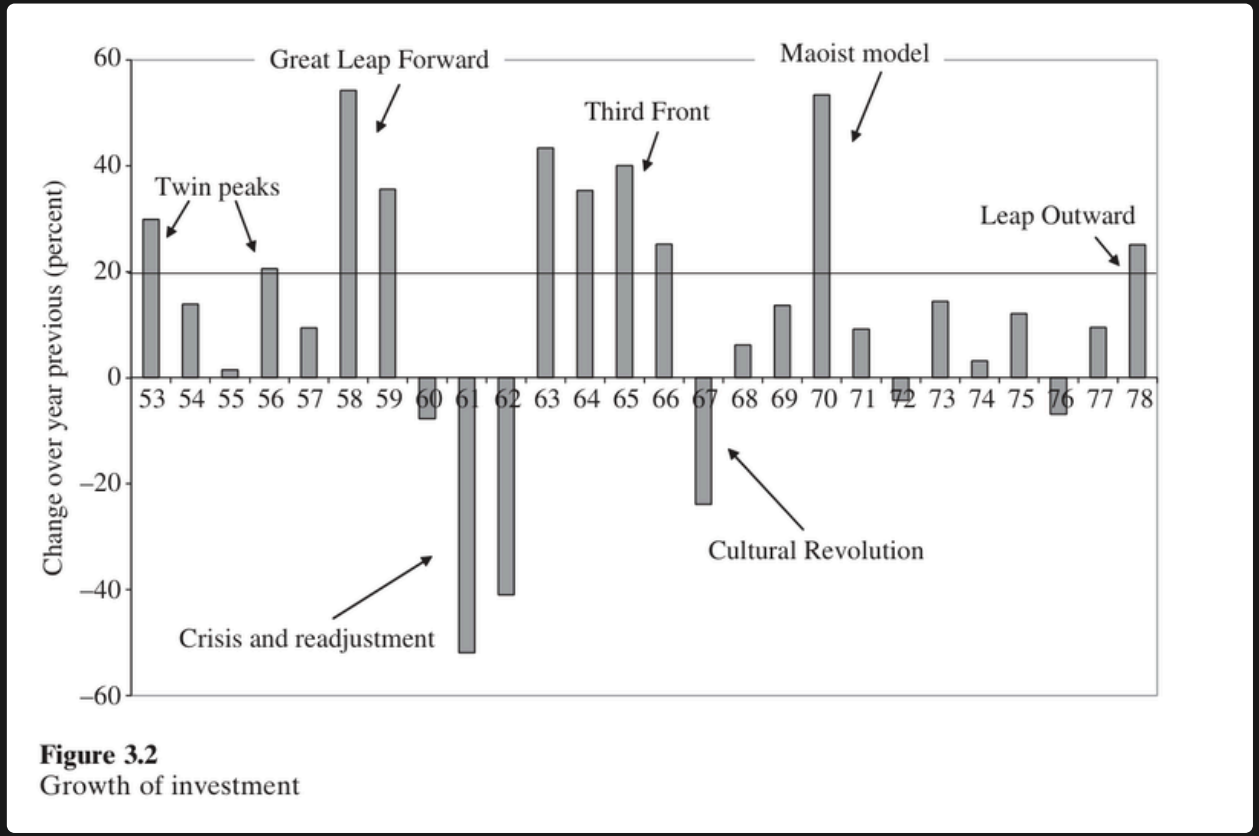


Figure 3.2
Growth of investment

Naughton (2007), *The Chinese Economy*, Fig. 3.2 — annual growth of investment

1958 — HUKOU

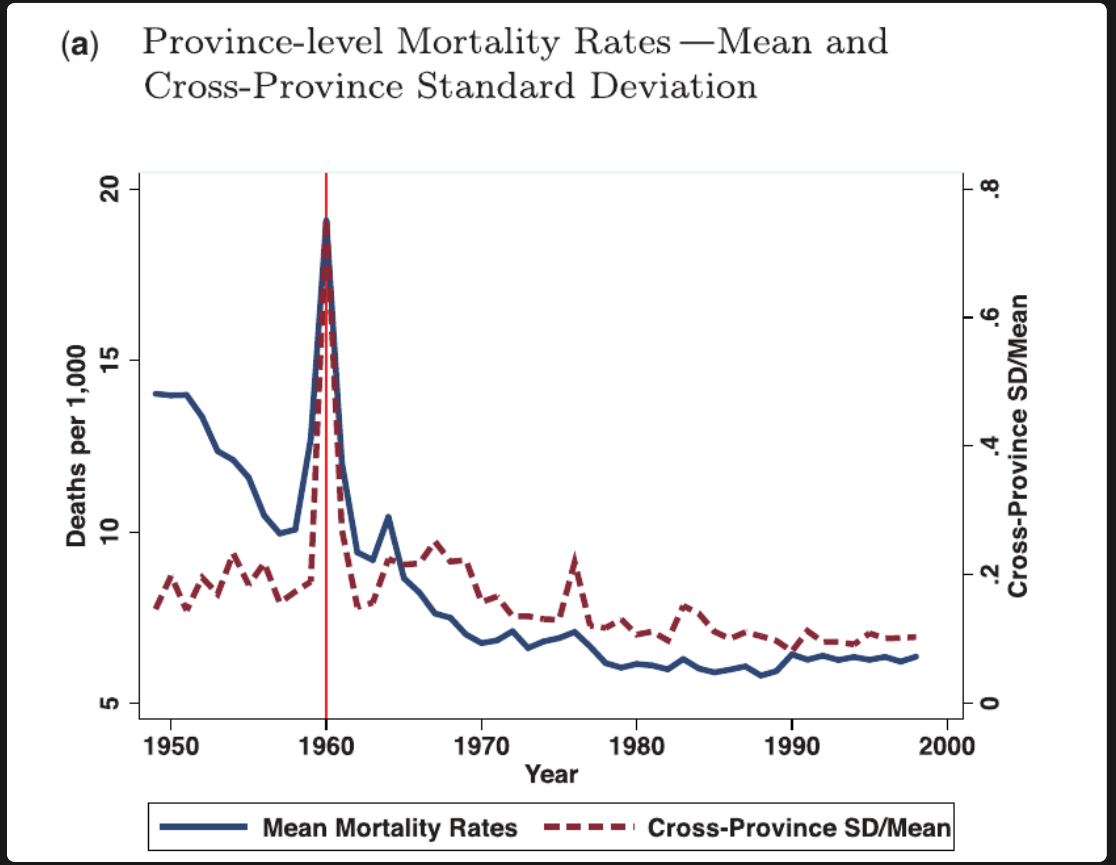
- **1955**: grain rationing begins (lasts over 30 years)
- **1958**: household registration formally instituted — working outside one's hukou place or category is prohibited
- After the Leap: urban registration a "meal ticket", rural registration "a license to starve"
- 1961-62: six million recent urban arrivals sent back to the villages; by the mid-1960s, a "caste system"

THE GREAT LEAP FORWARD, 1958-60

- **Goal: industrialise fast**
  - Everything at once, regardless of real resource constraints: "a vision, rather than a plan"
- **Organisation**: communes; bonuses and free markets abolished; decisions decentralised; "walking on two legs"
- **Two fateful decisions**: less labour and land for farming; more grain taken by the state
- **Why nobody braked**: inflated reports — the statistical system, the leaders' "eyes and ears", collapsed

HOW MANY DIED, AND WHAT "UNEVEN" MEANS

- Nobody counted. Estimates of excess deaths: **16.5 million** (Coale 1981), **30 million** (Banister 1987), **45 million** (Dikötter 2010)
- In 1960 the national average death rate jumps from about 10 to about **19 per 1,000**
- And so does the *spread* between provinces (dashed line): the same national shock, very different local outcomes
- That is what "uneven" means: not just more deaths, but more different deaths



Meng, Qian, Yared (2015), Fig. 1a

THE PUZZLE

In 1959-61, average food availability in the Chinese countryside was enough to feed the population.

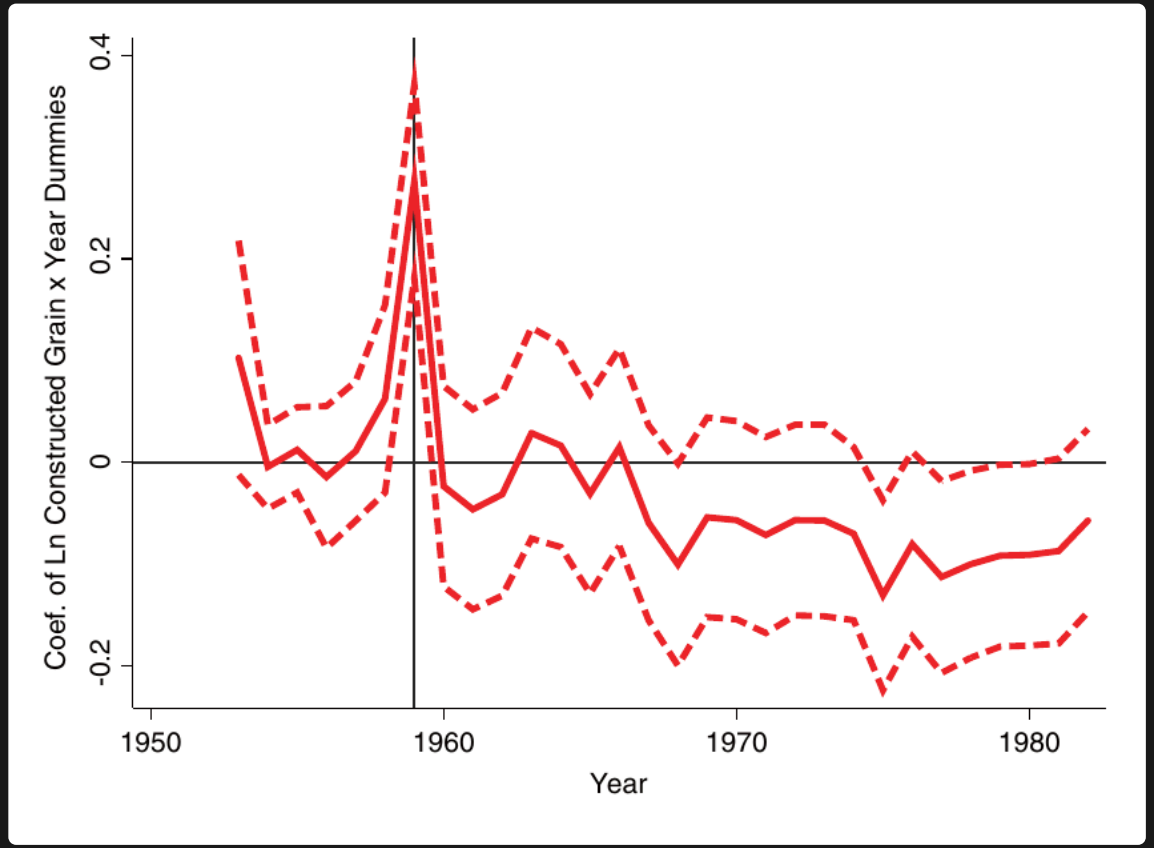
So why did 16.5 to 45 million people die?

- After the state took its share, the rural average in 1959 was about **2,300 kcal a day**; the survival benchmark is about **800**
- Production per person fell about 15% in 1959 — the largest fall of the period — but stayed far above that line

Meng, Qian, Yared (2015), *Review of Economic Studies* — 19 provinces, 77% of the population

THE SURPRISE

- We would expect more deaths where there was *less* food
- Across provinces, the paper finds the opposite: **more deaths where predicted grain output per head was higher** — only in the famine years
- How to read the graph: each point is a year; the line says by how much mortality differs between provinces whose output differs by 1% (a coefficient). Dashed lines: 95% interval
- About zero in normal years; clearly positive in 1958-59



Meng, Qian, Yared (2015), Fig. 3

HOW DO WE MEASURE THIS WITH BAD DATA?

- Official figures for output and deaths are unreliable in these years
- **Proxy 1**: output *predicted* from weather, geography and rural population, not the reported one
- **Proxy 2**: the missing children. In the 1990 census, how large is each birth cohort, in 511 counties?
- A small cohort born in 1959-61 = more deaths and fewer births. And the counties with better land lose the most cohorts (right)

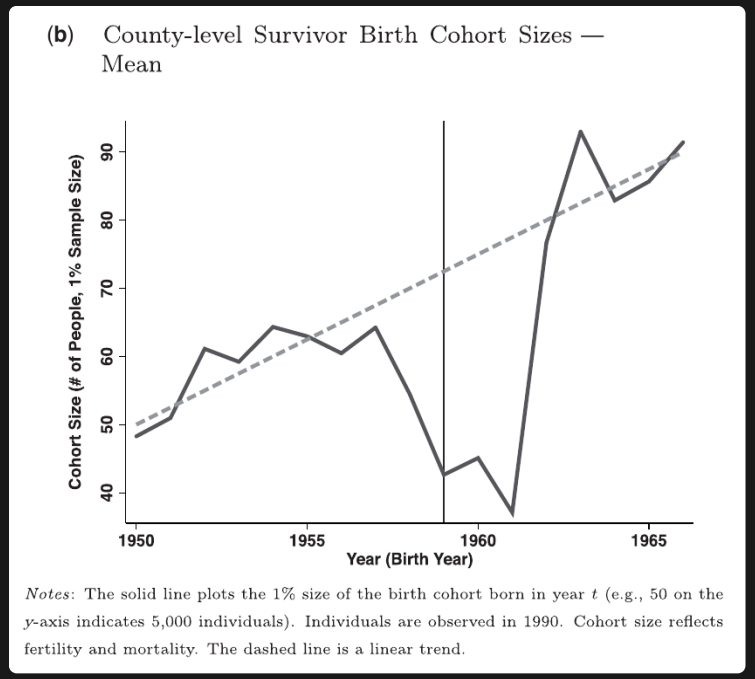


Fig. 1b: cohort size, by birth year

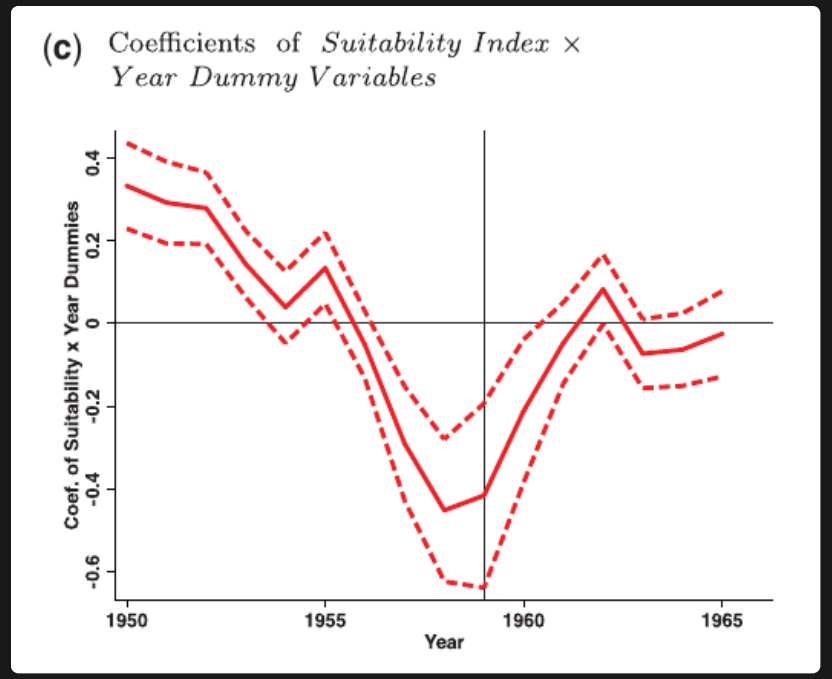
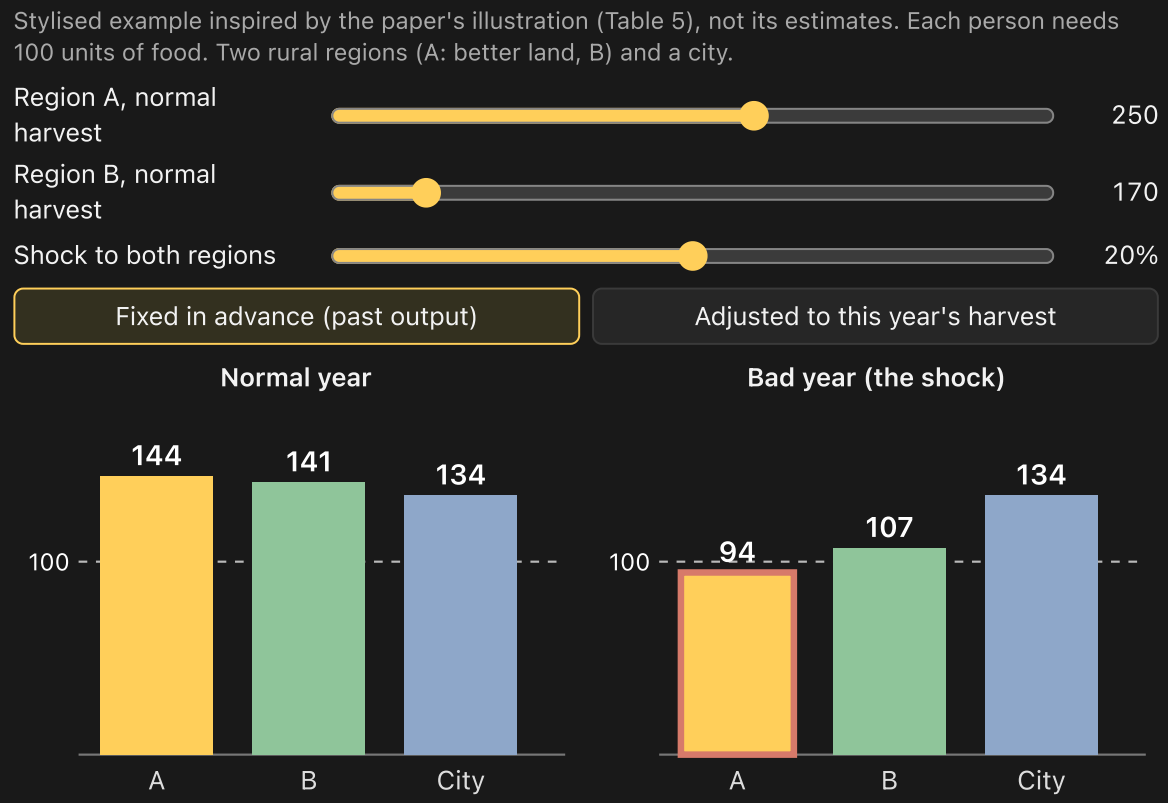


Fig. 4c: land suitability x birth year

HOW THE STATE COLLECTED GRAIN: A TOY ECONOMY

- Quotas were set **in advance**, from past verified output, top-down (province, prefecture, county)
- Each collective delivered its quota **even if the harvest fell short**
- Regions with higher output faced a **higher share** taken
- Try it: with the default numbers, after the shock region A (better land) is left with **less** than region B



WHAT THE PAPER ACTUALLY ESTIMATES

- **Unit**: a province in a year; 19 provinces, 1953-1982. **Outcome**: log deaths the following year
- **Regressor**: log grain output per head, *predicted* from weather, geography, rural population and land (not the reported figure)
- **Question**: do provinces with more output per head have more deaths, and does this change in the famine years (1958-60)?
- Year effects absorb everything that hits all provinces alike; we control for total and urban population
- **Normal years**: α ≈ 0. **Famine years**: α + β > 0 (about 0.14)

(1) province, famine vs normal years — Table 2

$$m_{p,t+1} = \alpha P_{p,t} + \beta\, P_{p,t} I_t^{Fam} + Z'_{p,t}\gamma + \delta_t + \varepsilon_{p,t}$$

(2) one coefficient per year — Figure 3

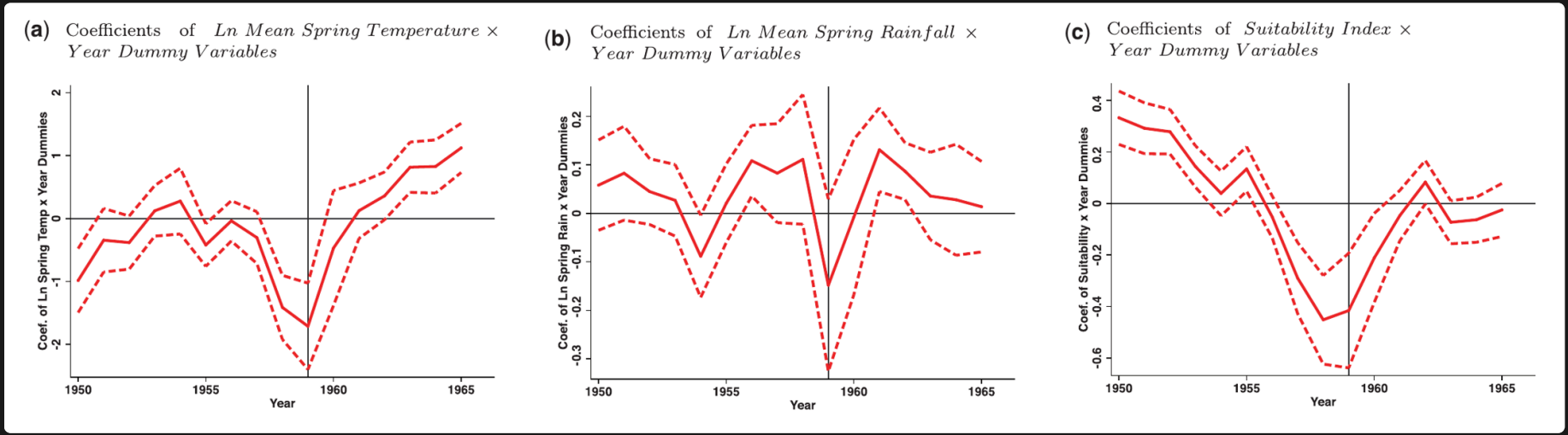
$$m_{p,t+1} = \sum_\tau \alpha_\tau P_{p,t} I_t^\tau + Z'_{p,t}\gamma + \delta_t + \varepsilon_{p,t}$$

(2′) counties, birth cohorts — Figure 4 (rebuilt from the text)

$$\ln C_{c,t+1} = \sum_\tau \left(a_\tau T_{c,t} + b_\tau R_{c,t} + s_\tau S_c\right) I_t^\tau + \gamma \ln \bar{C}_c + \delta_{p(c)} + \delta_t + \varepsilon_{c,t}$$

m: log deaths · P: log predicted grain output per head · I^{Fam}: 1 in 1958-60 · I^τ: year dummy · Z: log total and urban population · δ: fixed effects
C: size of the birth cohort (1990 census) · T, R: log spring temperature, rainfall · S: grain suitability · $\bar{C}$: county average cohort

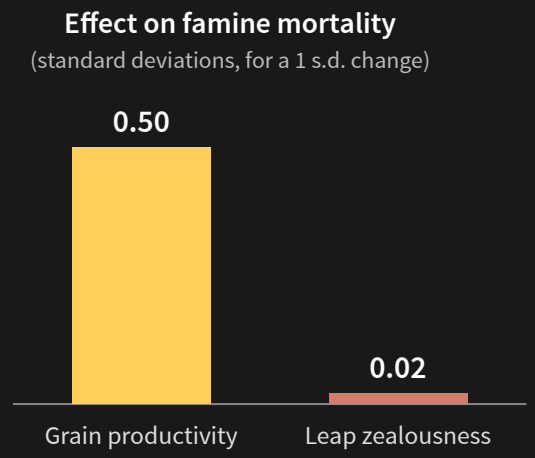
FIGURE 4: THE SAME QUESTION, BY COUNTY



Meng, Qian, Yared (2015), Fig. 4. Coefficients of (a) spring temperature, (b) spring rainfall and (c) land suitability, each interacted with birth-year dummies, with 95% intervals; outcome: log size of the birth cohort, 511 counties

IS IT JUST GREAT LEAP ZEAL? AND HOW BIG IS IT?

- Robustness of equation (1). Alternative: Great Leap **zealousness** (sown-area cuts, steel output, communal dining, purges, late liberation), urban bias, misreported output
- First check, in 17 provinces: were more productive provinces more zealous? If anything, *less*
- Add the zeal measure (their first principal component) and its interaction with the famine dummy to equation (1): the result barely changes
- Back-of-the-envelope: the procurement mechanism accounts for roughly **one third to 43%** of famine mortality — which leaves room for the Leap's policies and for politics



1961-76 — READJUSTMENT, THIRD FRONT, CULTURAL REVOLUTION

- **1961-63** readjustment: investment chopped back, about 20 million workers sent to the countryside
- **1964-66** Third Front: a new expansion into the interior
- **1966-76** Cultural Revolution: 17 million urban school-leavers sent to the countryside (*xiafang*); schools and universities closed
- **1976**: Mao dies

1976-78 — THE TURN

- Growth had accelerated under the Big Push — but even more in Taiwan and Hong Kong, far ahead of the PRC by 1978
- That demonstration effect convinced Chinese leaders to reform and open up
- **December 1978**: Third Plenum — ease the terms of trade with agriculture, "give farmers a chance to catch their breath"
- Deng Xiaoping consolidates power; *gaige kaifang*, reform and opening

A QUESTION, FOR NEXT WEEK

The famine paper argues that fixed procurement quotas, and not just the Leap's policies, made the famine deadlier where there had been more food. ***Imagine you could run any experiment you like.*** *What would you change, in which places, and what would you measure? And what is the thing you still could not observe?*

- Also: pick **one number** from today that you would not trust. Describe the **ideal data**: what would be measured, for whom, by whom, and what would still be missing
- And: what was the least clear point in today's lecture?
- Open answers, anonymous, ungraded

scan with your phone, or open:
docs.google.com/forms/d/e/1FAIpQLSdJ-sN-UY0IVRH3Mkkj20xz_2E9v7TsZBKiGN1arnVtdgiyfw/viewform

Answer before the next lecture. → Lecture 2: how do we answer a question like this?